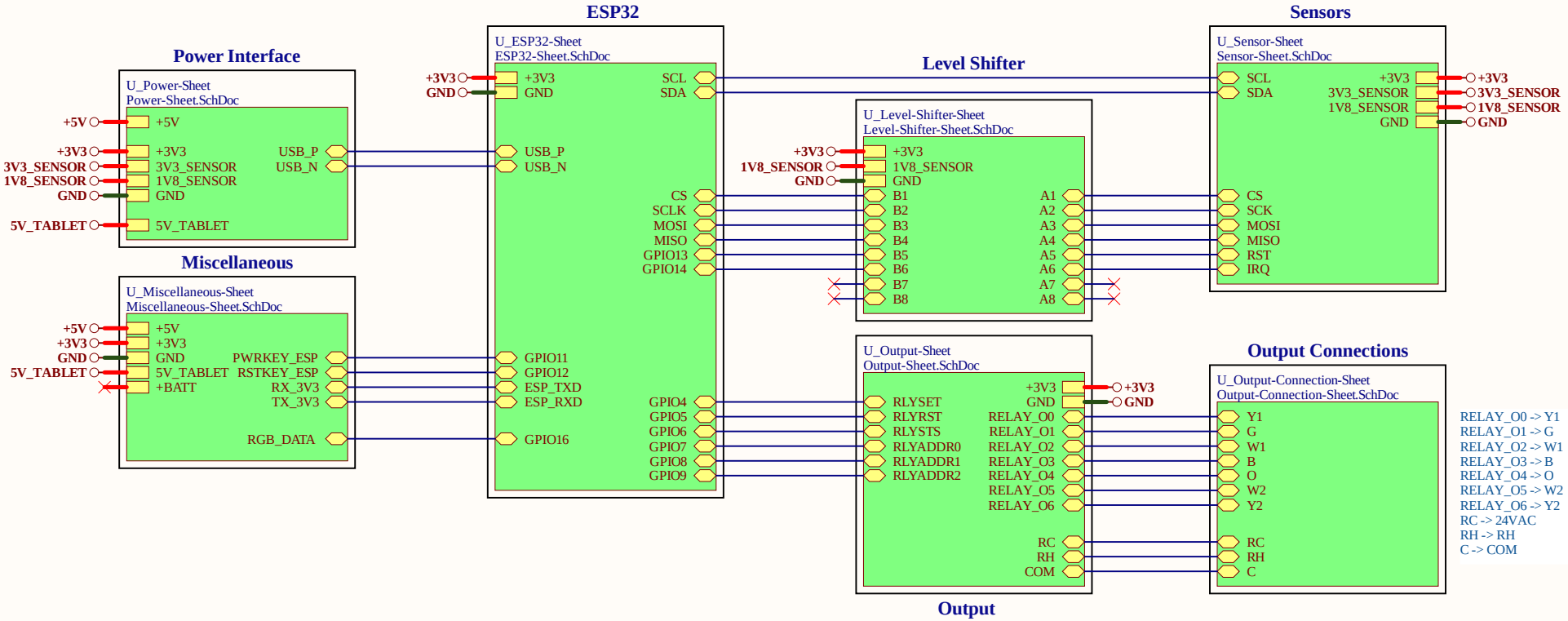


Relays, depending on the channel, switch between COM and 24VAC or RH wire.

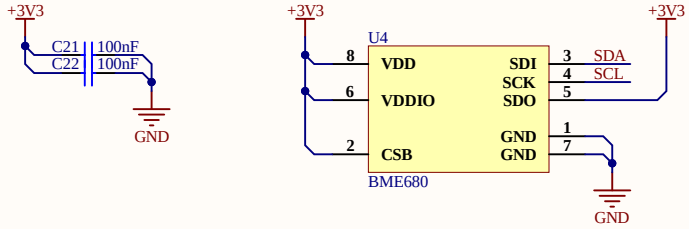
Y, G, O are switched betwen COM and 24VAC (RC)
W, B are switched between COM and RH wire

If C-wire is absent, 24VAC is absent, only W and B are switched to RH wire.

Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Top-Sheet.SchDoc	
File: ezlo-thermostat.PrjPcb	
Date: 1/7/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 9

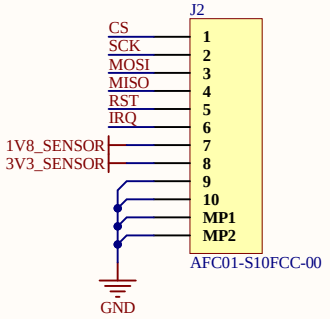
4-in-1 VOC Humidity Pressure Temperature



CSB -> VDDIO = I2C active
CSB -> GND = SPI active
I2C ADDR = BIN 1110111 HEX 0x77
I2C pull-up resistors are on the ESP32 sheet.

BSEC: Bosch Sensortec Environmental Cluster
BSEC precisely performs several calculations outside the device such as ambient air temperature, ambient relative humidity, pressure and air quality (IAQ) level.
https://www.bosch-sensortec.com/software-tools/software/bsec/
https://github.com/BoschSensortec/BSEC-Arduino-library

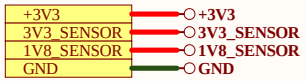
Radar Sensor



Radar sensor sits externally on it's own PCB
and is connected via FPC to the maimboard.



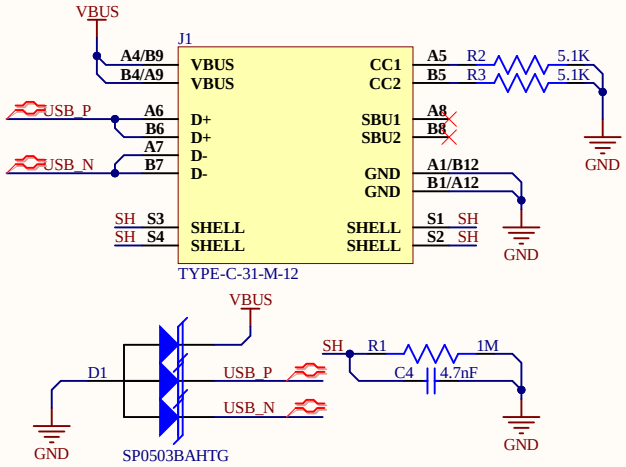




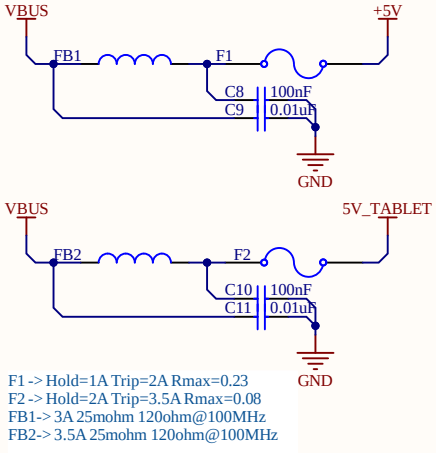
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Sensor-Sheet.SchDoc	
File: ezlo-thermostat.PrjPcb	
Date: 1/7/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 9

USB

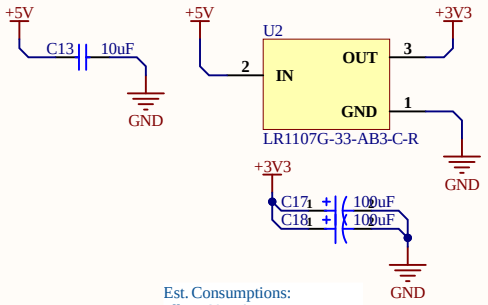


Filters



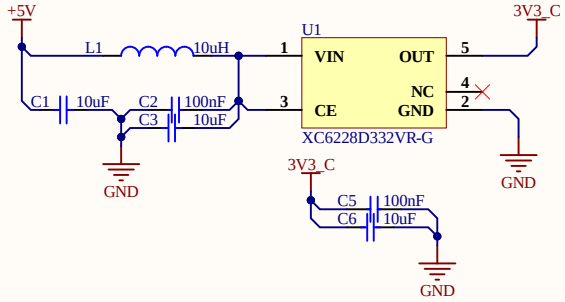
F1 -> Hold=1A Trip=2A Rmax=0.23
F2 -> Hold=2A Trip=3.5A Rmax=0.08
FB1-> 3A 25mohm 120ohm@100MHz
FB2-> 3.5A 25mohm 120ohm@100MHz

LDO 3.3V

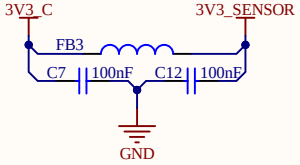


Est. Consumptions:
Idle: 160 mA
Max: 1050mA (worst case!)

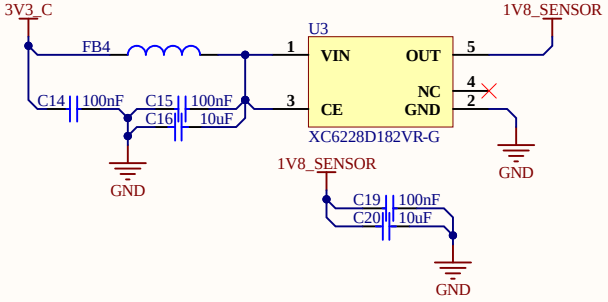
LDO 3.3V (Radar Sensor)



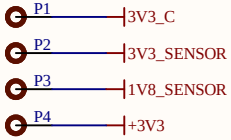
Filter

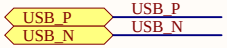


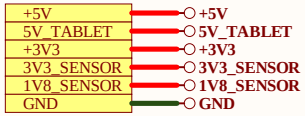
LDO 1.8V



Test Points

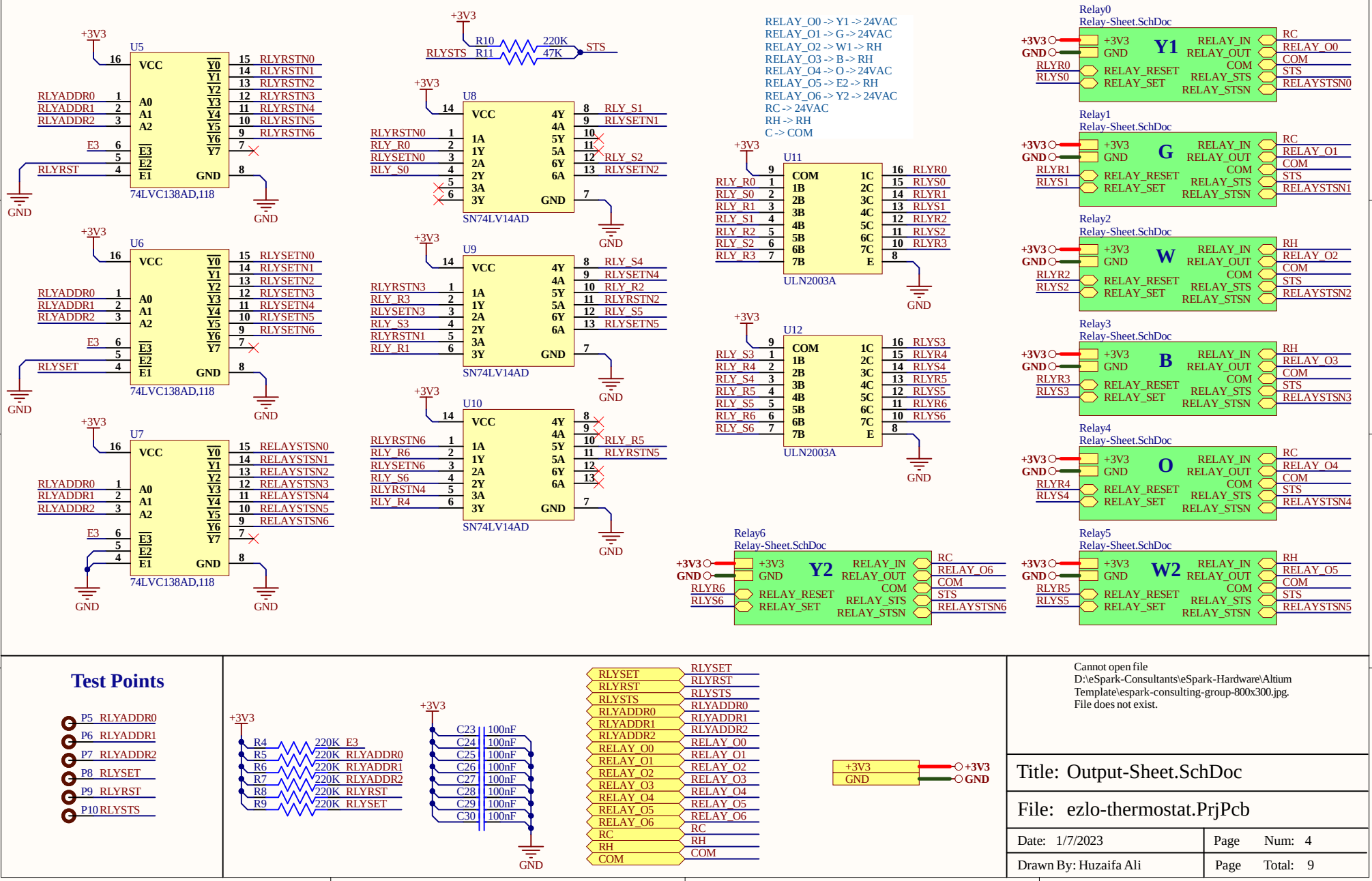


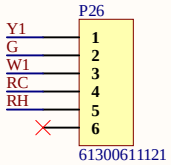




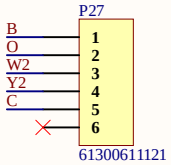
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Power-Sheet.SchDoc	
File: ezlo-thermostat.PrjPcb	
Date: 1/7/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 9

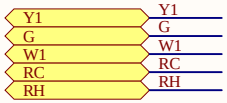

U5
74LVC138AD,118
U6
74LVC138AD,118
U7
74LVC138AD,118
U8
SN74LV14AD
U9
SN74LV14AD
U10
SN74LV14AD
U11
ULN2003A
U12
ULN2003A
RELAY_O0 -> Y1 -> 24VAC
RELAY_O1 -> G -> 24VAC
RELAY_O2 -> W1 -> RH
RELAY_O3 -> B -> RH
RELAY_O4 -> O -> 24VAC
RELAY_O5 -> E2 -> RH
RELAY_O6 -> Y2 -> 24VAC
RC -> 24VAC
RH -> RH
C -> COM
Relay0
Relay-Sheet.SchDoc
Y1
Relay1
Relay-Sheet.SchDoc
G
Relay2
Relay-Sheet.SchDoc
W
Relay3
Relay-Sheet.SchDoc
B
Relay4
Relay-Sheet.SchDoc
O
Relay5
Relay-Sheet.SchDoc
W2
Relay6
Relay-Sheet.SchDoc
Y2
Test Points
P5 RLYADDR0
P6 RLYADDR1
P7 RLYADDR2
P8 RLYSET
P9 RLYRST
P10 RLYSTS
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium
Template\espark-consulting-group-800x300.jpg.
File does not exist.
Title: Output-Sheet.SchDoc
File: ezlo-thermostat.PrjPcb
Date: 1/7/2023
Page Num: 4
Drawn By: Huzaifa Ali
Page Total: 9

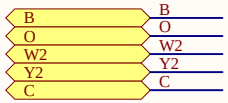


Need part# from client



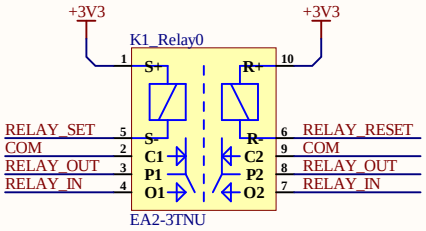
RELAY_O0 -> Y1
RELAY_O1 -> G
RELAY_O2 -> W1
RELAY_O3 -> B
RELAY_O4 -> O
RELAY_O5 -> W2
RELAY_O6 -> Y2
RC -> 24VAC
RH -> RH
C -> COM





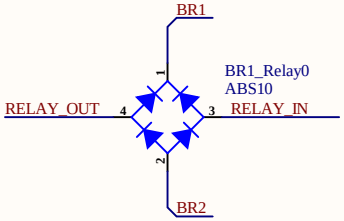
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Output-Connection-Sheet.SchDoc	
File: ezlo-thermostat.PrjPcb	
Date: 1/7/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 9

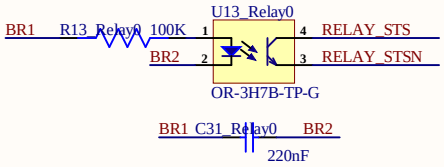




Connect RELAY_IN to 24VAC or RH AC Input

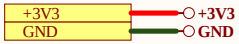


Add one 3V3 pullup externally to RELAY_STS







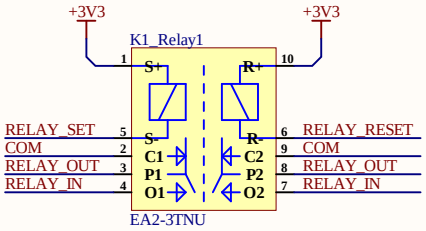


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Relay-Sheet.SchDoc

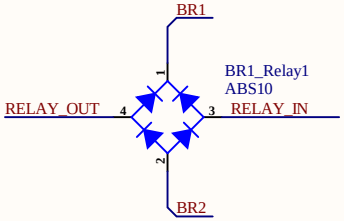
File: ezlo-thermostat.PrjPcb

Date: 1/7/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

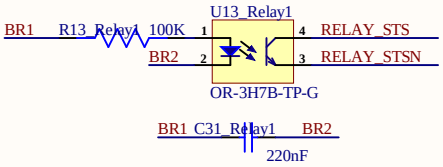




Connect RELAY_IN to 24VAC or RH AC Input

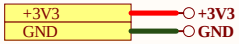


Add one 3V3 pullup externally to RELAY_STS







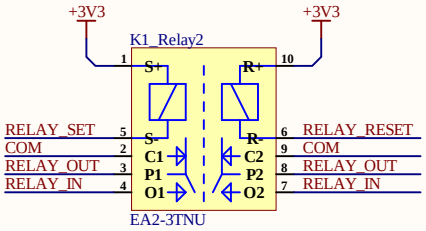


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Relay-Sheet.SchDoc

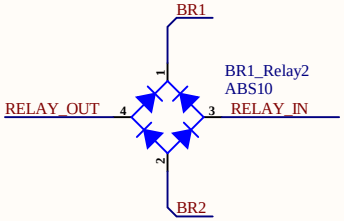
File: ezlo-thermostat.PrjPcb

Date: 1/7/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

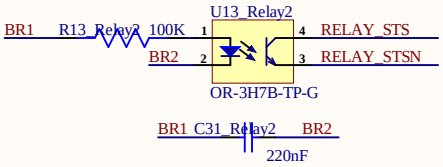




Connect RELAY_IN to 24VAC or RH AC Input

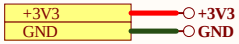


Add one 3V3 pullup externally to RELAY_STS







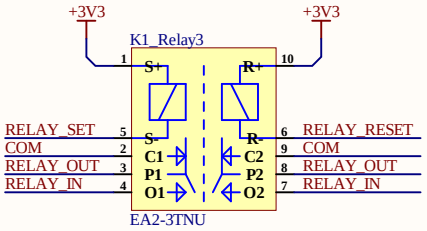


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Relay-Sheet.SchDoc

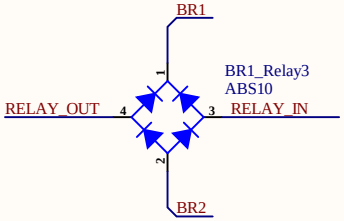
File: ezlo-thermostat.PrjPcb

Date: 1/7/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

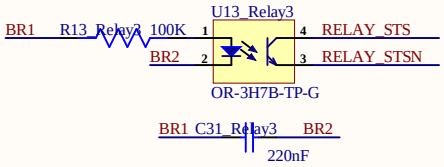




Connect RELAY_IN to 24VAC or RH AC Input

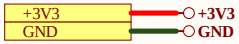


Add one 3V3 pullup externally to RELAY_STS







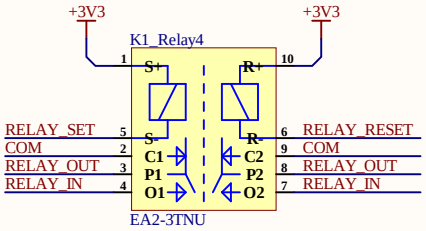


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Relay-Sheet.SchDoc

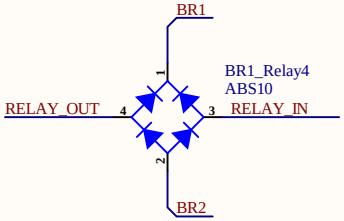
File: ezlo-thermostat.PrjPcb

Date: 1/7/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

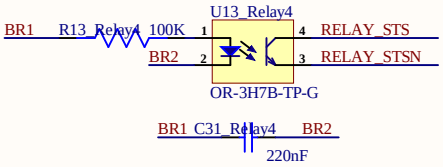




Connect RELAY_IN to 24VAC or RH AC Input

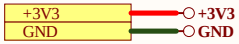


Add one 3V3 pullup externally to RELAY_STS







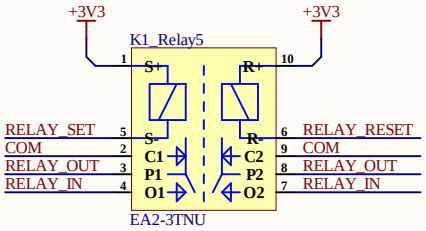


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Relay-Sheet.SchDoc

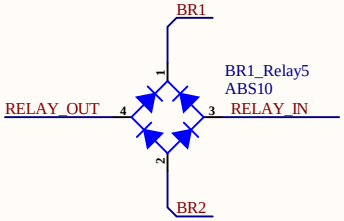
File: ezlo-thermostat.PrjPcb

Date: 1/7/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

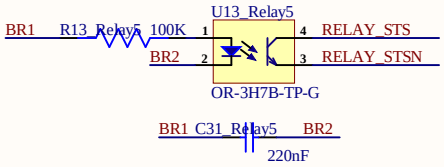




Connect RELAY_IN to 24VAC or RH AC Input

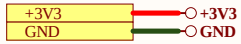


Add one 3V3 pullup externally to RELAY_STS







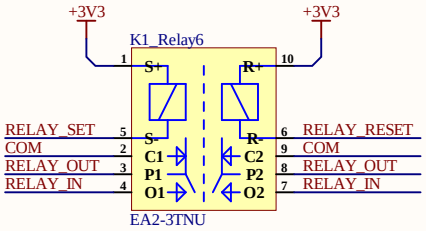


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Relay-Sheet.SchDoc

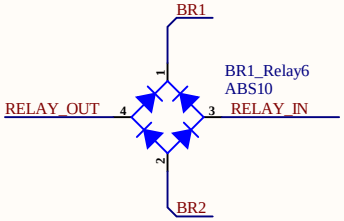
File: ezlo-thermostat.PrjPcb

Date: 1/7/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

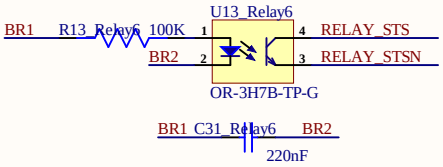




Connect RELAY_IN to 24VAC or RH AC Input

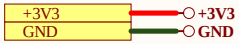


Add one 3V3 pullup externally to RELAY_STS









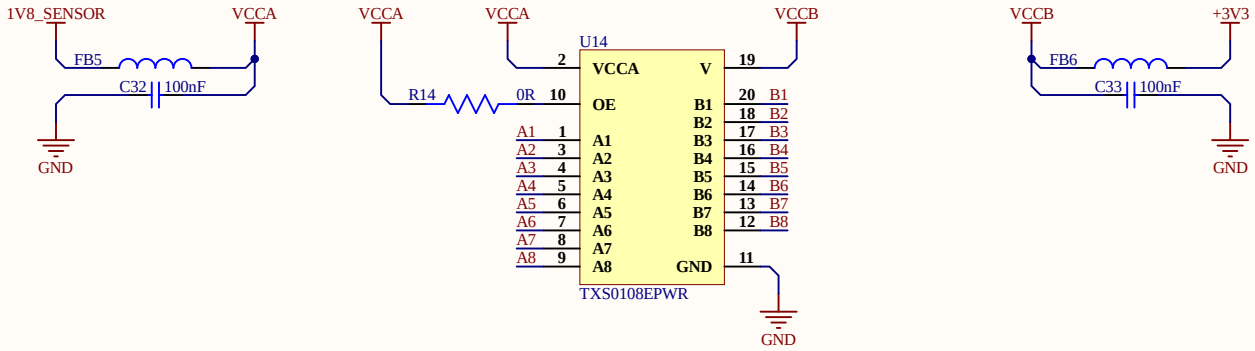
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Relay-Sheet.SchDoc

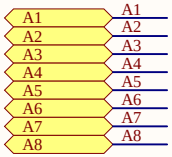
File: ezlo-thermostat.PrjPcb

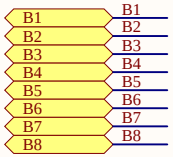
Date: 1/7/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

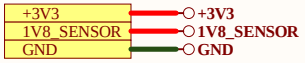
Level Shifter



1.4V to 3.6V on A port and 1.65V to 5.5V on B port (VCCA <= VCCB)!



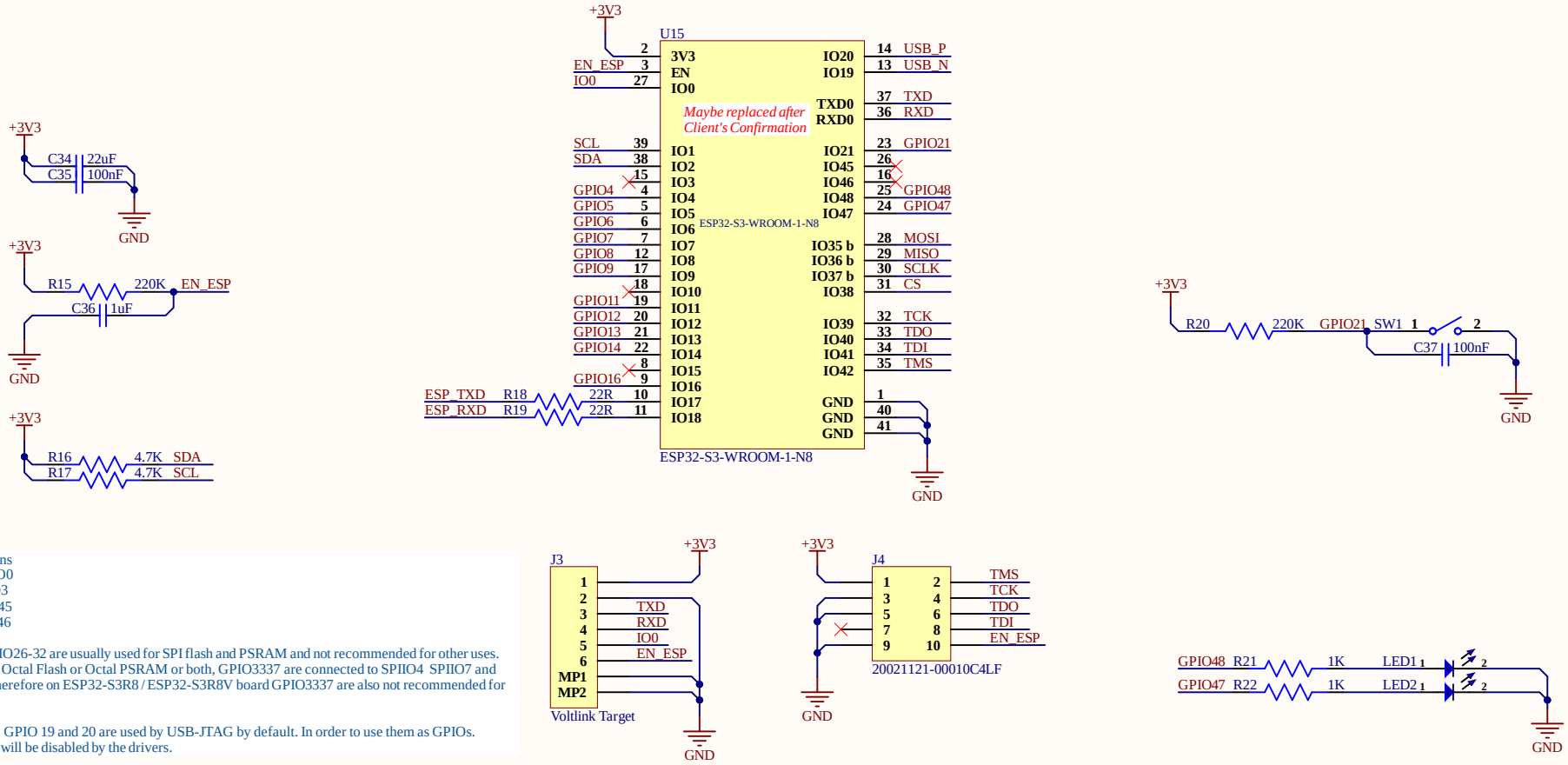




Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Level-Shifter-Sheet.SchDoc	
File: ezlo-thermostat.PrjPcb	
Date: 1/7/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 9

ESP32 MCU

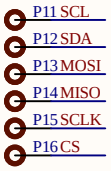


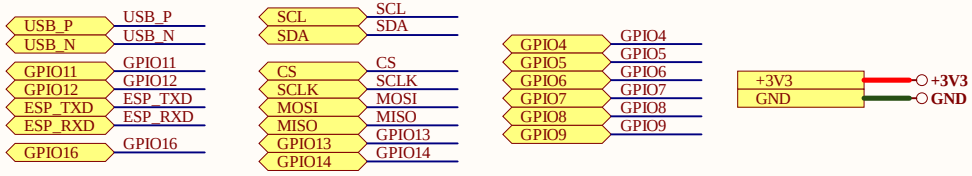
Strapping pins
- BOOT = IO0
- JTAG = IO3
- VSPI = IO45
- LOG = IO46

SPI0/1: GPIO26-32 are usually used for SPI flash and PSRAM and not recommended for other uses. When using Octal Flash or Octal PSRAM or both, GPIO3337 are connected to SPIIO4 SPIIO7 and SPIDQS. Therefore on ESP32-S3R8 / ESP32-S3R8V board GPIO3337 are also not recommended for other uses.

USB-JTAG: GPIO 19 and 20 are used by USB-JTAG by default. In order to use them as GPIOs. USB-JTAG will be disabled by the drivers.

Test Points

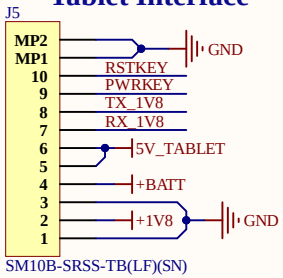




Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: ESP32-Sheet.SchDoc	
File: ezlo-thermostat.PrjPcb	
Date: 1/7/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 9

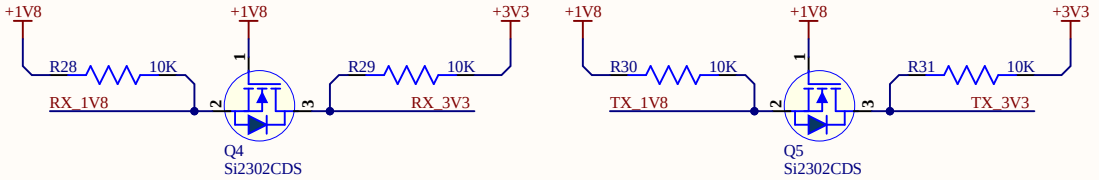
Tablet Interface



*interface with tablet 10 pins JST-SH 1A 50VDC rating, pinout contains:
10. RESETKEY (output going to tablet, pulled high by tablet to 1V8, need to be pulled down via mosfet to GND)
9. PWRKEY (output going to tablet, pulled high by tablet to VBAT, needs to be pulled down via mosfet by ESP32)
8. TX (with voltage translation 1V8 > 3V3 so that ESP32 can talk to tablet processor)
7. RX (with voltage translation 1V8 > 3V3 so that ESP32 can talk to tablet processor)
6. VUSB (output going to tablet USB for power & charging)
5. VUSB (output going to tablet USB for power & charging)
4. VBAT (input coming from battery)
3. GND
2. 1V8 (input coming from tablet)
1. GND

Battery on tablet is not needed so VBAT is not used in this revision!

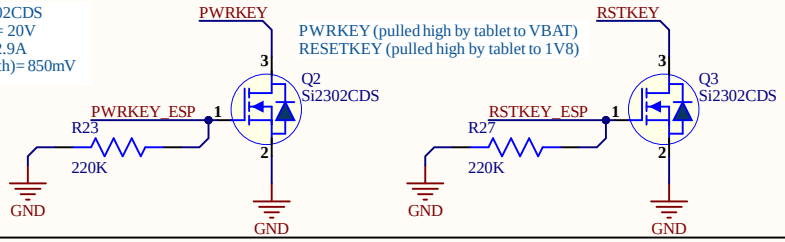
UART Level Conversion



If a UART connection between Android tablet and ESP32 is not needed parts of this schematic will be DNP!

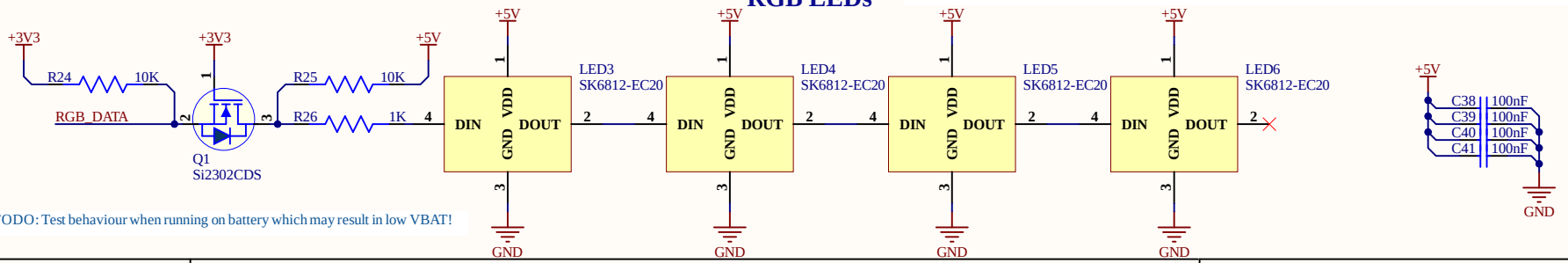
Tablet PWR & RST Keys

Si2302CDS
Vds = 20V
Id = 2.9A
Vgs(th)= 850mV



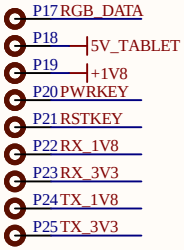
RGB LEDs

Max current per individual SK6812 at max brightness is 40mA. Total max current for 4 x SK6812 is 160mA on the +5V rail.

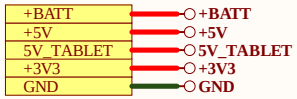


TODO: Test behaviour when running on battery which may result in low VBAT!

Test Points







Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Miscellaneous-Sheet.SchDoc	
File: ezlo-thermostat.PrjPcb	
Date: 1/7/2023	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 9